

Chapter 5: Food Safety

Textbook Exercise Solutions & Answers



1. Complete the following statements by using the correct option from those given below.

(Irradiation, dehydration, pasteurization, natural, chemical)

(a) Drying the food grains from farms under the hot sun is called

dehydration

(b) Materials like milk are instantly cooled after heating up to a certain high temperature. This method of food preservation is called

pasteurization

(c) Salt is a type of food preservative.

natural

(d) Vinegar is a type of food preservative.

chemical

2. Answer the following questions in your own words.

(a) How is milk pasteurised?

Boil the milk at 80°C for 15 minutes and cool it quickly. This destroys the microbes present in the milk and it can remain for a longer duration. This process is called pasteurization of milk.

(b) Why should we not consume adulterated food materials?

Different types of adulterants affect our health in different ways. Some adulterants cause abdominal discomfort or poisoning while some may affect the functioning of some organs if consumed over a long period of time or even cause cancer.

(c) What precautions do your parents take to keep foodstuffs safe?

Our parents take following care to keep foodstuffs safe

1. Drying of grains.
2. Boiling of milk, soups and curries from time to time.
3. Refrigeration of vegetables, fruits, milk and cooked food.
4. Candying of jams.
5. Use natural preservatives like oil, spices, neem leaves, salt, etc.
6. Use chemical preservatives in sauces, ketchups, pickles, jams and squashes.

(d) How does food spoilage occur? Which are the various factors spoiling the food?

Food spoilage is the process in which food deteriorates to the point in which it is not edible to humans or its quality of edibility becomes reduced.

Following are the factors for spoiling of food:

1. Bacteria causes food to spoil
2. Incorrect storage may spoil the food.
3. Infestation by pests.
4. Chemical reaction takes place in food and it gets spoiled.

(e) Which methods of food preservation would you use?

I use following methods to preserve the food:

1. Freezing
2. Boiling
3. Salting of pickles
4. drying of grains
5. candying of jams.

3. What shall we do?

(a) There are vendors selling uncovered sweet-meats in open places in the market.

We should tell them to sell the covered sweets because uncovered sweets are harmful to eat because it contains dust, dirt and germs. And buyers also should not purchase these uncovered sweets.

(b) A 'pani-puriwala' is serving the panipuri with dirty hands.

We should tell him to wear gloves before serving panipuri because dirty hands contain dirt and germs which are harmful to us.

(c) We have purchased a large quantity of fruits and vegetables.

We should keep them in refrigerator because in refrigerator, due to low temperature, fruits and vegetables do not spoil and biological and chemical reactions in fruits and vegetables are slowed down at low temperature.

(d) We need to protect foodstuffs from pests like rats, cockroaches, wall-lizards etc.

1. If we do not protect foodstuffs from pests like rats, cockroaches, wall-lizards etc. then the food get spoiled by them and germs carried by them enters into the food.
2. If we eat this food then we may get food poisoning and we fall sick so we need to protect foodstuffs from pests like rat, cockroaches, wall lizards etc.

4. Find the odd-man-out.

(a) Salt, vinegar, citric acid, sodium benzoate.

Salt

(b) Lakhi dal, brick dust, metanyl yellow, turmeric powder.

Turmeric powder

(c) Banana, apple, guava, almond.

Almond

(d) Storing, freezing, settling, drying

Storing

5. Complete the chart below.

Foodstuff	Adulterant
1. Turmeric powder	Metanil yellow
2. Black pepper	Dried papaya seeds
3. Rava	Iron filings
4. Honey	Sugar, water and jaggery

6. Explain why this happens and suggest possible remedies.

(a) Qualitative wastage of food.

Qualitative wastage of food happens due to wrong methods of protecting food, excessive use of food preservatives, over-cooking, washing the vegetables after cutting them, mis-handling of fruits like grapes and mangoes, miscalculation of the time required to transport food from producers to consumers are some of the reasons of qualitative wastage of food.

Possible remedies:

1. Avoid overcooking of food.
2. Store grains and other perishable foodstuffs like vegetables, fruits, milk etc. using proper methods.
3. Wash fruits and vegetables before cutting it.

(b) The cooked rice is underdone.

Sometimes in a hurry if we cook the rice it is underdone.

Possible remedies: Use proper pressure cooker to cook the rice.

(c) The wheat that was bought is a bit moist.

Sometimes due to sudden rain, wheat gets a bit moist. Possible remedies: Do not store the moist wheat, it gets spoiled due to fungus so first sundry it and then store in a clean and dry container to avoid microbial growth.

(d) The taste of yoghurt is too sour/slightly bitter.

The taste of yoghurt is too sour/slightly bitter means it is spoiled. This happens if it is not kept in the refrigerator.

Possible remedies:- Always keep the yoghurt in refrigerator to avoid biological and chemical reactions in food materials.

(e) Cut fruits turned black.

Fruit contains an enzyme called polyphenol oxidase or tyrosinase that reacts with oxygen. The oxidation reaction basically forms a sort of rust on the surface of fruits so it turns black.

Possible remedies:

1. Coat the fruits with sugar syrup
2. Add lemon juice on fruits.

7. Give reasons.

(a) Food remains safe at 5° Celsius.

Food remains safe at 5° Celsius because at 5°C, micro-organisms stop growing.

(b) Nowadays, food is served in buffet style during large gatherings.

Nowadays, food is served buffet style during large gatherings because due to buffet style quantitative wastage of food can be avoided, as people take only as much as they could eat.